

Similarly, the constancy of η and the fact that $\frac{\partial \eta_i}{\partial \mathbf{w}_1} = X$ reduces the second equation in (1) to:

$$\mathbb{P}(Z = i) \frac{\partial \mathcal{L}(\eta, i)}{\partial \eta_i} \cdot \mathbb{E}[X|Z = i] + \mathbb{P}(Z \neq i) \frac{\partial \mathcal{L}(\eta, \neq i)}{\partial \eta_i} \cdot \mathbb{E}[X|Z \neq i] = 0. \quad (3)$$

Solving for $\mathbb{P}(Z = i) \frac{\partial \mathcal{L}(\eta, i)}{\partial \eta_i}$ in (2) and substituting in (3) gives us:

$$\mathbb{P}(Z \neq i) \frac{\partial \mathcal{L}(\eta, \neq i)}{\partial \eta_i} \cdot \left(\mathbb{E}[X|Z \neq i] - \mathbb{E}[X|Z = i] \right) = 0.$$

If $\mathbb{P}(Z \neq i) = 0$, then $\mathbb{E}[X] = \mathbb{E}[X|Z = i]$ is trivially true. Otherwise, using the non-vanishingness of the off-category partial derivative $\frac{\partial \mathcal{L}(\eta, \neq i)}{\partial \eta_i}$, division yields the equivalence of $\mathbb{E}[X|Z = i]$ to $\mathbb{E}[X|Z \neq i]$, and hence to the unconditional mean $\mathbb{E}[X]$. $\square$

We now show that Lemma 3.2 applies to the widely used cross entropy loss:

Theorem 3.3. *If the class probabilities $\mathbb{P}(Z = j)$ are all nonzero, and the trivially obtainable loss is optimal when $\mathcal{L}(\eta, z) = -\log \frac{\exp(\eta_z)}{\sum_{i=1}^k \exp(\eta_i)}$, then each class has the same mean $\mathbb{E}[X|Z = z]$.*

Proof. In this case, the trivial predictor $\eta_\tau(Z)_j = \log(\mathbb{P}(Z = j))$ exists, achieving the trivially obtainable loss, which we have assumed optimal. Furthermore, $\mathcal{L}$ has on-category partial derivative $\partial \mathcal{L}(\eta, i)/\partial \eta_i = \exp(\eta_i)/\sum_{j=1}^k \exp(\eta_j) - 1 \in (-1, 0]$, and nonvanishing off-category partial derivative $\partial \mathcal{L}(\eta, \neq i)/\partial \eta_i = \exp(\eta_i)/\sum_{j=1}^k \exp(\eta_j) \in (0, 1)$, both bounded, so the conditions of Lemma 3.2 apply. $\square$

3.3 Linearly Guarded Labels Have Zero Covariance with the Features

The next theorem establishes the equivalence of conditions 3 and 4.

Theorem 3.4. *Let X be a random vector taking values in $\mathbb{R}^d$ with finite first moment, and Z a random vector taking values in $\{0, 1\}^k$ with one-hot encoding, with each class probability $\mathbb{P}(Z = j)$ being nonzero. Then the class-conditional means $\mathbb{E}[X|Z = j]$ are all equal to the unconditional mean $\mathbb{E}[X]$ if and only if every component of X has zero covariance with every component of Z , i.e. the cross-covariance matrix Σ_{XZ} , whose $(i, j)^{\text{th}}$ entry is $\text{Cov}(X_i, Z_j)$, is the zero matrix.*

Proof. Since Z is one-hot, we can rewrite the $(i, j)^{\text{th}}$ entry of Σ_{XZ} as:

$$\mathbb{E}[X_i Z_j] - \mathbb{E}[X_i] \mathbb{E}[Z_j] = \mathbb{P}(Z = j) \left(\mathbb{E}[X_i|Z = j] - \mathbb{E}[X_i] \right).$$

As $\mathbb{P}(Z = j) > 0$, it follows that $\mathbb{E}[X_i|Z = j] = \mathbb{E}[X_i]$ if and only if $\text{Cov}(X_i, Z_j) = 0$. $\square$

We have thus established the equivalence of the first four conditions stated earlier. See Appendix C for the last one, on statistical parity.

4 Least-Squares Concept Erasure

In Section 3 we saw that X linearly guards Z if and only if each component of X has zero covariance with each component of Z . We will now characterize the set of affine transformations $r(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbf{b}$ such that $r(X)$ linearly guards Z .

Theorem 4.1. *Let X and Z be random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively, with X of finite first moment. Then given some affine function $r(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbf{b}$, the modified random vector $r(X)$ linearly guards Z if and only if the columns of the cross-covariance matrix Σ_{XZ} are contained in the null space of $\mathbf{P}$.*

Proof. From Theorem 3.4 we know that $r(X)$ linearly guards Z if and only if $\text{Cov}(r(X), Z)$ is the zero matrix. By the linearity property of cross-covariance, we have:

$$\text{Cov}(r(X), Z) = \text{Cov}(\mathbf{P}X + \mathbf{b}, Z) = \mathbf{P}\text{Cov}(X, Z) = \mathbf{P}\Sigma_{XZ}.$$

Therefore, $r(X)$ linearly guards Z if and only if $\ker(\mathbf{P}) \supseteq \text{colsp}(\Sigma_{XZ})$. $\square$

Implications for prior work. Notably, the above theorems imply that three previously proposed methods in the literature, Spectral Attribute Removal (SAL) [36], Mean Projection [17], and Fair PCA [20], are guaranteed to achieve linear guardedness given suitable hyperparameters. See Appendix D for further discussion.

4.1 Derivation of LEACE

Theorem 4.1 is a very weak condition, which is far from identifying unique values for $\mathbf{P}$ and $\mathbf{b}$. In most applications, however, we’d like to make a “small” edit to $\mathbf{X}$ so that useful information contained in $\mathbf{X}$ is maximally preserved. We operationalize the notion of a small edit in terms of the mean squared norm $\mathbb{E}\|\mathbf{r}(\mathbf{X}) - \mathbf{X}\|_{\mathbf{M}}^2$ defined by some positive-definite inner product $\mathbf{M}$,⁴ which can be thought of as a local quadratic approximation to *any* measure of divergence between $\mathbf{X}$ and $\mathbf{r}(\mathbf{X})$ (such as Kullback–Leibler divergence, for example). While we are primarily interested in the Euclidean ($\mathbf{M} = \mathbf{I}$) and Mahalanobis ($\mathbf{M} = \Sigma_{\mathbf{X}\mathbf{X}}^+$) norms, it will turn out that there is a *single* erasure function that minimizes *all* such norms simultaneously. We will see in Section 6 that ensuring edits are small in this sense provides substantial benefit to downstream task performance as compared to other methods which also guard the labels $\mathbf{Z}$.

Below, we derive the optimal eraser under the assumption that $\mathbf{X}$ and $\mathbf{Z}$ are centered.

Theorem 4.2. *Let $\mathbf{X}$ and $\mathbf{Z}$ be centered random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively, each of finite second moment. Let $\mathbf{M} \in \mathbb{R}^{d \times d}$ be a p.s.d. matrix defining a (possibly degenerate) inner product on $\mathbb{R}^d$: $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \mathbf{x}^T \mathbf{M} \mathbf{y}$. Let $\Sigma_{\mathbf{X}\mathbf{X}} \in \mathbb{R}^{d \times d}$ be $\mathbf{X}$ ’s covariance matrix, and $\Sigma_{\mathbf{X}\mathbf{Z}} \in \mathbb{R}^{d \times k}$ be the cross-covariance matrix of $\mathbf{X}$ and $\mathbf{Z}$. Let $\mathbf{A}^+$ denote the Moore–Penrose pseudoinverse of a matrix $\mathbf{A}$, and let $\mathbf{A}^{1/2}$ be the p.s.d. square root of a p.s.d. matrix $\mathbf{A}$. Then the objective*

$$\operatorname{argmin}_{\mathbf{P} \in \mathbb{R}^{d \times d}} \mathbb{E} \left[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|_{\mathbf{M}}^2 \right] \quad \text{subject to } \operatorname{Cov}(\mathbf{P}\mathbf{X}, \mathbf{Z}) = \mathbf{0}$$

has the following solution:

$$\mathbf{P}^* = \mathbf{I} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}}} \mathbf{W},$$

where $\mathbf{W}$ is the whitening transformation $(\Sigma_{\mathbf{X}\mathbf{X}}^{1/2})^+$ and $\mathbf{P}_{\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}}} = (\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}})(\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}})^+$ is the orthogonal projection matrix onto $\operatorname{colsp}(\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}})$.

Proof. See Appendices E.1 and E.2 for two independent proofs of Theorem 4.2. □

The above theorem assumes that the random vectors $\mathbf{X}$ and $\mathbf{Z}$ are centered, and does not include a bias term. Below we extend our results to the uncentered case, and derive the optimal bias $\mathbf{b}^*$.

Theorem 4.3. *Let $\mathbf{X}$ and $\mathbf{Z}$ be random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively, each of finite second moment. Define $\mathbf{M}$ and $\mathbf{P}^*$ as in Theorem 4.2 and $\mathbf{b}^* = \mathbb{E}[\mathbf{X}] - \mathbf{P}^* \mathbb{E}[\mathbf{X}]$. Then $(\mathbf{P}^*, \mathbf{b}^*)$ minimizes $\mathbb{E}\|\mathbf{P}\mathbf{X} + \mathbf{b} - \mathbf{X}\|_{\mathbf{M}}^2$, subject to $\operatorname{Cov}(\mathbf{P}\mathbf{X} + \mathbf{b}, \mathbf{Z}) = \mathbf{0}$.*

Proof. Let $\mathbf{P} \in \mathbb{R}^{d \times d}$ and define $\tilde{\mathbf{X}} = \mathbf{X} - \mathbb{E}[\mathbf{X}]$ and $\mathbf{c} = \mathbf{P}\mathbb{E}[\mathbf{X}] + \mathbf{b} - \mathbb{E}[\mathbf{X}]$. Then,

$$\begin{aligned} \mathbb{E}\|\mathbf{P}\mathbf{X} + \mathbf{b} - \mathbf{X}\|_{\mathbf{M}}^2 &= \mathbb{E}\|(\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}) + \mathbf{c}\|_{\mathbf{M}}^2 \\ &= \mathbb{E}\|\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\|_{\mathbf{M}}^2 + 2\mathbb{E}[\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}]^T \mathbf{M} \mathbf{c} + \mathbf{c}^T \mathbf{M} \mathbf{c} \\ &= \mathbb{E}\|\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\|_{\mathbf{M}}^2 + \mathbf{c}^T \mathbf{M} \mathbf{c}, \end{aligned}$$

where we have eliminated the middle term because $\mathbf{P}$ is linear and $\mathbb{E}[\tilde{\mathbf{X}}] = \mathbf{0}$. Since $\mathbf{M}$ is p.s.d., our objective is minimized for $\mathbf{c} = \mathbf{0}$, i.e. $\mathbf{b} = \mathbb{E}[\mathbf{X}] - \mathbf{P}\mathbb{E}[\mathbf{X}]$. The problem thus reduces to choosing $\mathbf{P}$ so as to minimize $\mathbb{E}\|\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\|_{\mathbf{M}}^2$ subject to $\operatorname{Cov}(\mathbf{P}\mathbf{X} + \mathbf{b}, \mathbf{Z}) = \operatorname{Cov}(\mathbf{P}\tilde{\mathbf{X}}, \mathbf{Z}) = \mathbf{0}$, which Theorem 4.2 shows occurs when $\mathbf{P} = \mathbf{P}^*$. □

⁴Our proofs also include degenerate “inner products” where $\mathbf{M}$ is singular, and the associated seminorms.

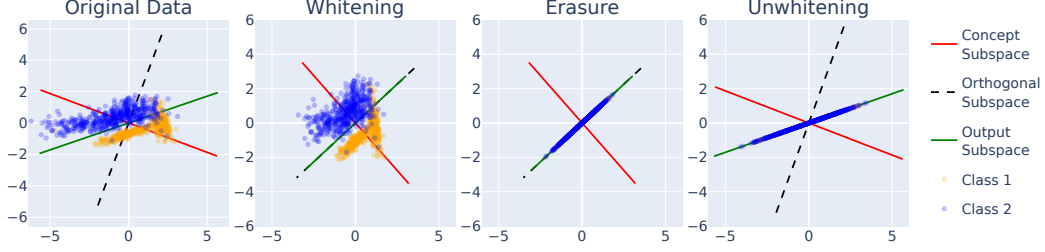


Figure 1: LEACE projection in 3 steps. First the data is whitened, ensuring equal variance in all directions. It is then orthogonally projected onto $\text{colsp}(\mathbf{W}\Sigma_{XZ})^\perp$, guaranteeing linear guardedness. Finally, we unwhiten the data so that its covariance structure mimics the original.

Putting together Theorems 4.2 and 4.3 and rearranging, we arrive at the LEACE formula:

$$r_{\text{LEACE}}(\mathbf{x}) = \mathbf{x} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W}(\mathbf{x} - \mathbb{E}[\mathbf{X}]) \quad (1)$$

Intuitively, LEACE de-means and whitens $\mathbf{x}$, projects onto the subspace responsible for correlations between X and Z , then unwhitens the result. Finally, it subtracts this value from $\mathbf{x}$, thereby surgically removing the linearly available information about Z .

4.2 Oblique Projections are Least-Squares Optimal

Prior work on linear concept erasure has assumed that erasure functions should be orthogonal projections [29, 32, 36], appealing to the well-known fact that an orthogonal projection of a point $\mathbf{x}$ onto a subspace U yields the nearest point in U to $\mathbf{x}$. But even in the case where X is centered, r_{LEACE} is *not* an orthogonal projection in general. Orthogonal projection matrices are symmetric, and $\mathbf{I} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W}$ is only symmetric in the special case where $\mathbf{P}_{\mathbf{W}\Sigma_{XZ}}$ and $\mathbf{W}$ commute. It is an *oblique* projection however, since applying $\mathbf{P}^*$ twice yields the same result as applying it once: $(\mathbf{P}^*)^2 = \mathbf{I} - 2\mathbf{W}\mathbf{P}_{\mathbf{W}\Sigma_{XZ}}\mathbf{W}^+ + \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W}\mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W} = \mathbf{P}^*$.

Orthogonal projections are generally not least-squares optimal for concept erasure because the necessary and sufficient condition for linear guardedness, $\mathbf{P}\Sigma_{XZ} = \mathbf{0}$, is a constraint on the *nullspace* of $\mathbf{P}$, and not on its range. We may freely choose the range of the projection to minimize the mean squared distance, as long as we zero out $\text{colsp}(\Sigma_{XZ})$. In Figure 1, an orthogonal projection would map all points onto the dashed line, thereby preserving less of the variance of the original data than LEACE does (green line). See Appendix F for a concrete example.

4.3 Extension to Continuous Z

While not a focus of this work, it’s worth noting that LEACE can also be applied to the setting where Z takes arbitrary values in $\mathbb{R}^k$, as long as we restrict ourselves to the ordinary least squares regression loss $\mathcal{L}(\eta, \mathbf{z}) = \|\eta - \mathbf{z}\|_2^2$. In particular, the proofs of equivalence between conditions 1 and 2 given in Appendix B make no categorical assumption on Z , and the equivalence between the optimality of a zero weight matrix (condition 2) and zero cross-covariance (condition 4) is well known in the OLS setting. We can then apply Theorems 4.2 and 4.3, which also make no categorical assumption, to derive the same optimal affine eraser as in the categorical case.

5 Evaluation

5.1 Intrinsic Evaluation

Following Ravfogel et al. [31] we evaluate the ability of our method to remove gender information from the last hidden layer of a frozen BERT model. We use the biographies dataset of De-Arteaga et al. [6], composed of short biographies annotated by both binary gender and profession. We embed each biography with the [CLS] embedding in the last layer of BERT, enforce the same-conditional-mean constraint to remove gender information from the [CLS], and then evaluate the performance of the model, after the intervention, on the main task of profession prediction. We